VITA / BIOGRAPHICAL SKETCH

Harrison Katz

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Education

- **Doctor of Philosophy (Ph.D.) in Statistics,** University of California, Los Angeles (UCLA)
  Dissertation: 'Advancing Bayesian Forecasting: A Bayesian Dirichlet Auto-Regressive Conditional Heteroskedasticity Model, a Bayesian Dirichlet Auto-Regressive Moving-Average Model, and Other Innovations.'
  Advisors: Robert E. Weiss (Co-Chair), Ying Nian Wu (Co-Chair)
- **Master of Science (M.S.) in Mathematics & Statistics,** Georgetown University
- **Bachelor of Arts (B.A.) in Political Science,** University of California, Los Angeles (UCLA)

Professional Experience

- **Senior Tech Lead Manager, Finance Data Science & Strategy, Airbnb,** 2026–present
  Lead the team owning top-line forecasting across global corridors, feeding earnings, treasury, and strategic planning.
- **Tech Lead Manager, Forecasting Data Science, Airbnb,** 2025–2026
  Managed a team of data scientists developing Bayesian forecasting methods for Finance; served as technical advisor to VP Finance on forecasting strategy across 21 country groups.
- **Staff Data Scientist, Airbnb,** 2024–2025
  Extended B-DARMA framework to handle time-varying volatility (B-DARCH); led production automation of forecasting models across thousands of time series.
- **Senior Data Scientist, Airbnb,** 2019–2024
  Lead developer of company-wide forecast models (e.g., Nights Booked, Revenue); created the Bayesian Compositional Time-Series framework (B-DARMA, B-DARCH) to forecast lead times during COVID-19.
- **Research Analyst (Risk Analysis), Federal Reserve,** 2016–2018
  Conducted rigorous data analytics, mathematical modeling, and research on credit-default-swap markets.

Publications

Peer-Reviewed Journal Articles

1. Katz, H. (2026). "Coupled Supply and Demand Forecasting in Platform Accommodation Markets." *Tourism Management*, 119, 105514.
2. Katz, H. (2026). "Centered-Innovation MA for Bayesian Dirichlet ARMA: Theoretical Equivalence and an Application to Bank-Asset Shares." *Journal of Forecasting*.
3. Katz, H. (2026). "Forecasting the Evolving Composition of Guest Origin Markets in Platform Bookings: A Bayesian Compositional Time-Series Approach Using Airbnb Data." *Forecasting*, 8(4), 74.
4. Katz, H. (2026). "Directional-Shift Dirichlet ARMA Models for Compositional Time Series with Structural Break Intervention." *International Journal of Forecasting*.
5. Katz, H. (2026). "Retraining as Approximate Bayesian Inference." *Foresight: The International Journal of Applied Forecasting*, Issue 81.

6. Katz, H., and R. E. Weiss (2026). "A Bayesian Dirichlet Auto-Regressive Conditional Heteroskedasticity Model for Forecasting Currency Shares." *International Journal of Forecasting*, 42(3), 1033–1046.
7. Katz, H., and E. Savage (2025). "Slomads Rising: Structural Shifts in U.S. Airbnb Stay Lengths During and After the Pandemic (2019–2024)." *Tourism & Hospitality* (MDPI), 6(4), 182.
8. Katz, H., and T. Maierhofer (2025). "Forecasting the U.S. Renewable-Energy Mix with a Bayesian Dirichlet ARMA Model." *Renewable Energy Forecasting: Innovations and Breakthroughs* (MDPI).
9. Katz, H., L. Medina, and R. E. Weiss (2025). "Sensitivity Analysis of Priors in the Bayesian Dirichlet Auto-Regressive Moving Average Model." *Forecasting*, 7(3), 32.
10. Katz, H., P. Coles, and E. Savage (2025). "Lead Times in Flux: Analyzing Airbnb Booking Dynamics During Global Upheavals (2018–2022)." *Annals of Tourism Research: Empirical Insights*.
11. Katz, H., E. Savage, and K. T. Brusch (2025). "Two-Part Forecasting for Time-Shifted Metrics." *Foresight: The International Journal of Applied Forecasting*, 77, 26–33.
12. Katz, H., K. T. Brusch, and R. E. Weiss (2024). "A Bayesian Dirichlet Auto-Regressive Moving Average Model for Forecasting Lead Times." *International Journal of Forecasting*, 40(4), 1556–1567.

Under Review

13. Katz, H. (2026). "Re-Specification as Approximate Bayesian Inference." Under review, *Foresight: The International Journal of Applied Forecasting*.
14. Katz, H., J. Needleman, and L. Medina (2026). "Distributional Fitting and Tail Analysis of Lead-Time Compositions: Nights vs. Revenue on Airbnb." Under review, *Annals of Tourism Research: Empirical Insights*.

Working Papers and Technical Notes

15. Katz, H. (2026). "Four Forecasts, One Word: Optimality Criteria, Their Incompatibilities, and the Governance of Prediction in Organizations." Working paper. SSRN 7380898.
16. Katz, H. (2026). "When Should Forecasting Models Be Re-Specified? A Cost-Sensitive Trigger for Adaptive Model-Form Updating." Technical note, arXiv:2606.06670.
17. Katz, H. (2026). "Cost-Sensitive Retraining via Posterior Learning Debt." Working paper. arXiv:2604.06438.
18. Katz, H. (2025). "Impact by Design: Translating Lead Times in Flux into an R Handbook." Technical note, arXiv:2511.12763.

Selected Presentations

- "Forecast & the Street." Fireside with Andrew Slabin, VP of Investor Relations, **Airbnb Strategic Finance & Analytics Offsite**, San Francisco, CA, July 2026.
- "A Bayesian Dirichlet Framework for Compositional Time Series Forecasting." Contributed paper, **46th International Symposium on Forecasting**, Montreal, Canada, June 28–July 1, 2026.
- "Forecasting in the Wild." Practitioner talk, **North Carolina State University**, Raleigh, NC, April 2026.
- "Forecasting Dynamics at Airbnb." Invited seminar, **UCLA Fielding School of Public Health,** Master of Data Science in Health (MDSH) Seminar Series, November 2024.
- **Guest Lecturer, Data Analysis (MBA).** UCLA Anderson School of Management, 2024–2025. Two lectures on applied forecasting methods.
- **Internal Technical Talks.** Airbnb, 2019–present. Multiple presentations on B-DARMA, B-DARCH, and two-part forecasting methods to cross-functional teams including Finance, Product, and Engineering.

Media and Public Writing

- "A Forecast Is One Word Doing Four Jobs." Interview, **Significance magazine** (Royal Statistical Society and American Statistical Association), Vol. 23, Issue 6, November 2026.
- "Forecasting Airbnb: Where Demand, Supply, and Uncertainty Collide." Guest, **IBF On Demand podcast** (Institute of Business Forecasting & Planning), hosted by Eric Wilson, September 2026.
- "When Forecasting Isn't About Accuracy." Guest, **Forecasting Impact podcast** (International Institute of Forecasters), June 2026.
- "How We Knew COVID Was Over (and What Our Models Had to Unlearn)." **Airbnb Tech Blog**, August 2026.
- "When History Fails You, Borrow from Geography." **Airbnb Tech Blog**, May 2026.
- "What COVID Did to Our Forecasting Models (and What We Built to Handle the Next Shock)." **Airbnb Tech Blog**, March 2026.